

المحاضره الثانيه

What is an algorithm?

- ❖ In computer programming, an algorithm is a set of well-defined instructions to solve a particular problem.
- ❖ It takes a set of input and produces a desired output.

Step 1: Start

1. بداية

Step 2: Define the variables

2. تعريف المتغيرات

Step 3: Read values of variables

3. قراءة قيمة كل متغير

Step 4: Process

4. معالجة

- math operations
- logic operations
- comparisons

▪ عمليات رياضية

▪ عمليات منطقية

▪ مقارنات

Step 5: Display result

5. عرض/طباعة الناتج

Step 6: End

6. نهاية

Algorithm to add two numbers

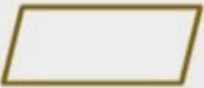
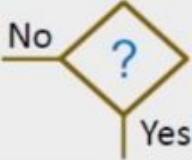
- ❖ We need to first write the steps (sequence of actions) that lead to performing the task.
- ❖ For example, an algorithm to add two numbers:
 - Take two number inputs
 - Add numbers using the + operator
 - Display the result

Step #	Description
Step1	Start
Step2	Declare or define variables <code>num1, num2, sum;</code>
Step3	Read values num1 and num2
Step4	Add num1 and num2 and assign the results to sum <code>sum ← num1 + num2;</code>
Step5	Print sum
Step6	End

Flowchart Symbols

- A Flowchart is a graphical representation that shows the behavior (workflow) of an algorithm.

Flowcharts use standard shapes including the following:

Symbol	Name	Description
	Start / End	An oval represents a start or end point
	Arrows	A line is a connector that shows flow direction between the representative shapes
	Input /Output	A parallelogram represents input or output
	Process	A rectangular represents a process (calculation)
	Decision	A diamond indicates a decision (comparison)
	Connector	A circle is used to combine one part of the flowchart with another part

Adding two numbers

✓ Write the pseudo-code for algorithm to sum two numbers. Then draw the equivalent flowchart.

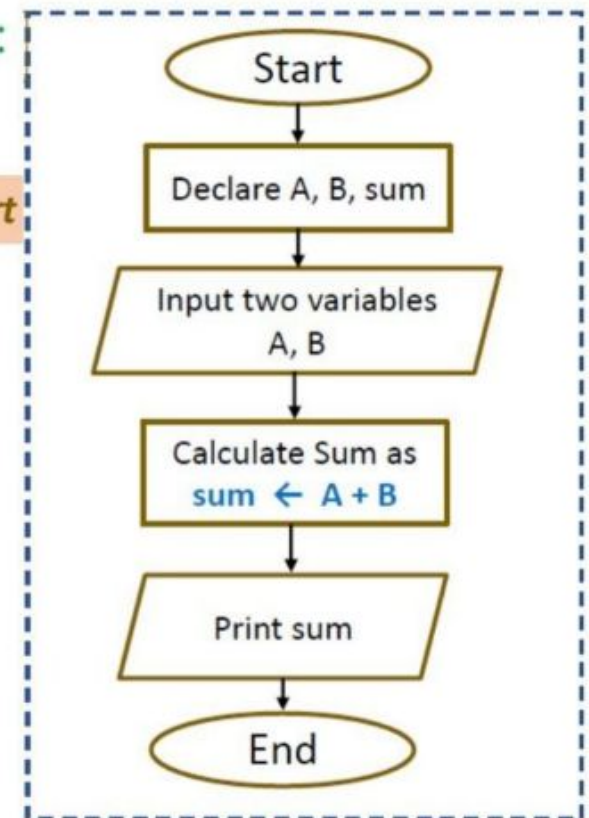
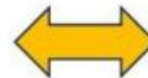
➤ Flowcharts use standard shapes including the following:

Ans:

Pseudo-code Algorithm

Step 1: START
Step 2: DECLARE variables num1, num2, sum;
Step 3: READ **variables** num1, num2;
Step 4: CALCULATE sum
 $\text{sum} = \text{num1} + \text{num2};$
Step 5: PRINT sum
Step 6: END

Flowchart

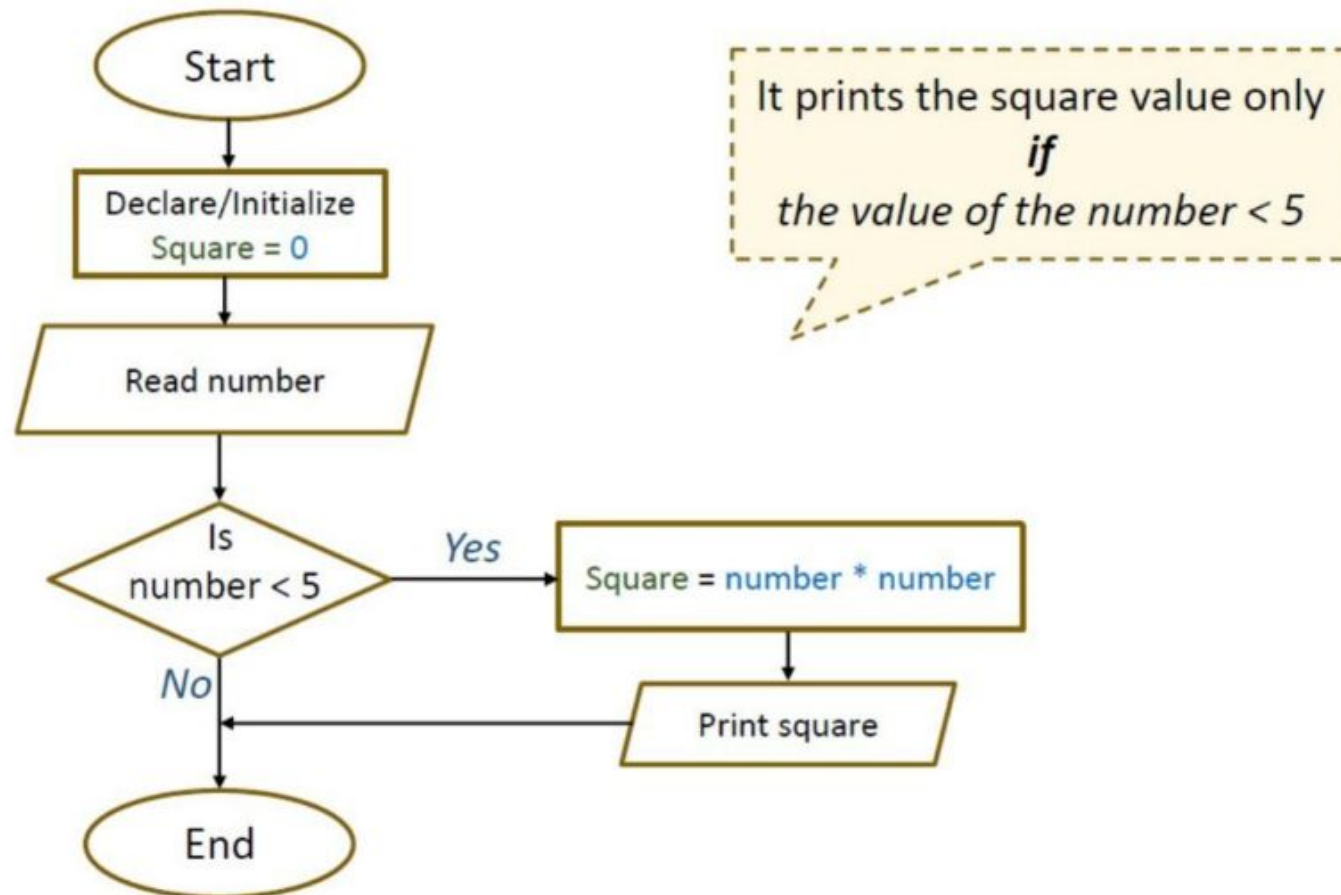


Note:

- The **pseudo-code** and **flowchart** of an algorithm is not written for a particular programming language.
- They can be used to plan a solution before coding it.

Print if the condition met

- ❖ Draw the Flowchart to print the square of a number if the number is less than 5.



Note:

- The lines and arrows in it show the sequence of steps and the relationship among them.

Difference between pseudo-code and flowchart

Main differences between Pseudo-code and Flowchart

- ❖ The pseudo-code is a high-level description of an algorithm while the flowchart is a graphical representation of an algorithm.
- ❖ An algorithm is a set of instructions for solving a problem or accomplishing a task.
- ❖ Every computerized device uses algorithms, which cut the time required to do things manually.

- ❖ There are three main ways or scenarios used in the programming. These are:
 - 1) **Sequencing** – putting things in the right order
 - 2) **Conditions** – performing different things depending on some rules
 - 3) **Iteration (Looping)** – repeating operations

Sequencing

- ❖ Calculate the average of three input numbers.

Pseudo-code Algorithm

Step 1: Start

Step 2: Declare variables Ave;

Step 3: Read variables num1, num2, num3;

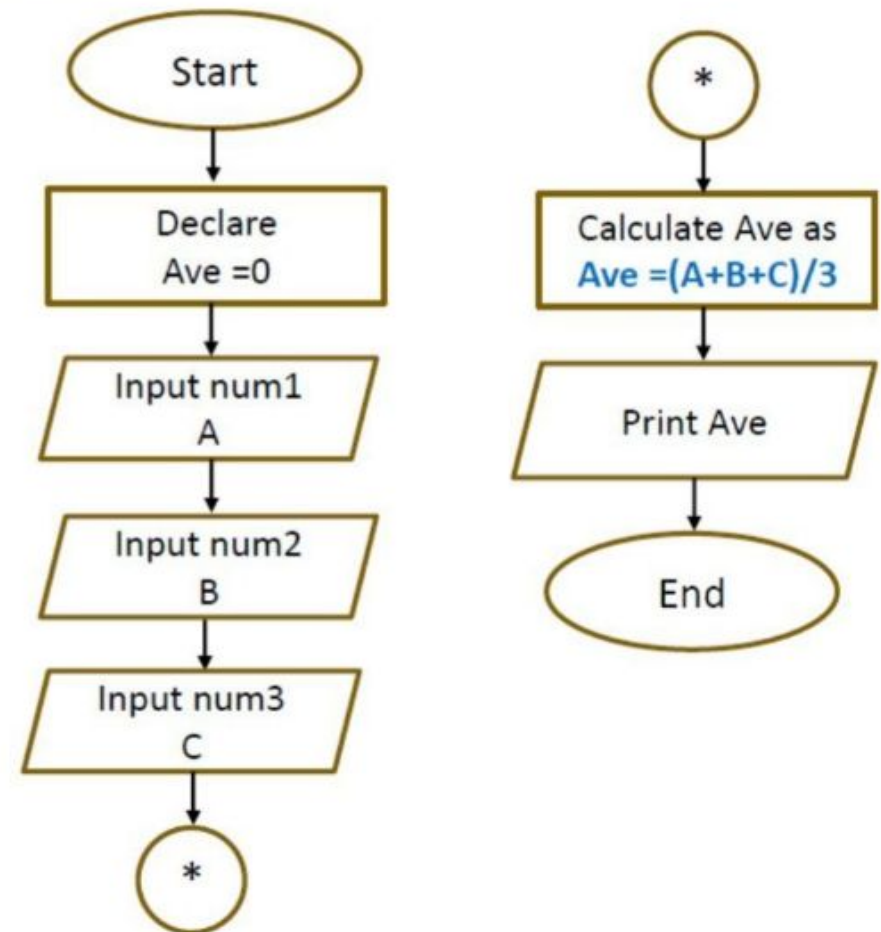
Step 4: Calculate Ave

$$\text{Ave} = (\text{num1} + \text{num2} + \text{num3})/3;$$

Step 5: Print Ave

Step 6: End

Flowchart



Conditions

- ❖ Conditions are very important to understand the flow-control of the code being executed.
- ❖ Conditions are used to make a decision (choose between two or more alternatives based on the condition)
- For example, conditions become very essential to use when facing different directions, so based on the condition should choose the thing to be executed.

Use the *if* statement

```
If (condition) then  
    run this statement;
```

```
Else  
    run the other statement;
```



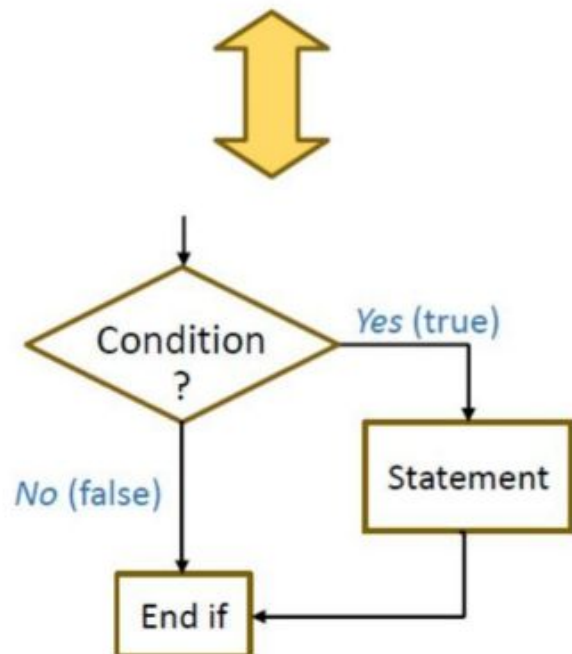
Conditions

1

if then

If (condition) then
statement(s)

End if



2

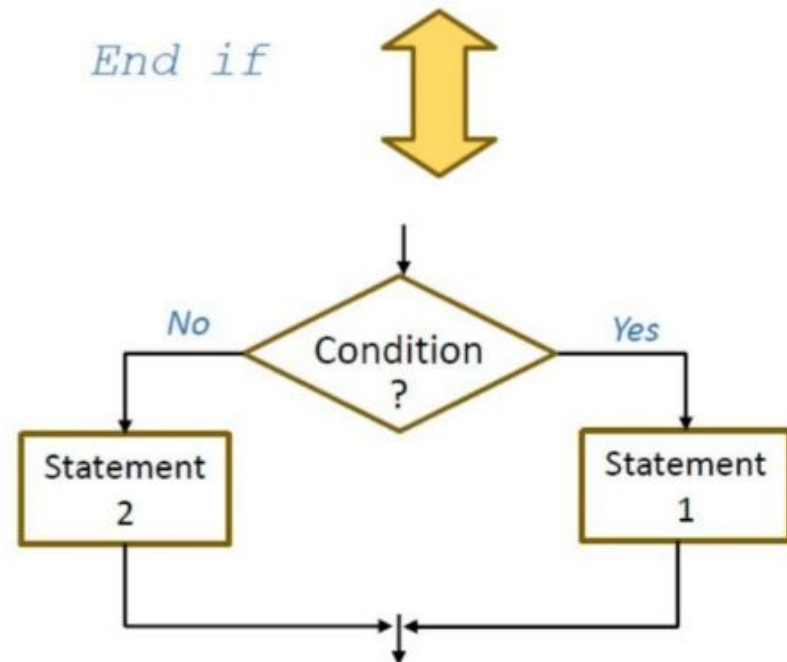
if then else

If (condition) then
statement 1

Else

statement 2

End if



Nested if structure

3

Nested structure of

if then

If (condition 1) then
statement 1

If (condition 2) then
statement 2

If (condition 3) then
statement 3

Else
statement 4

End if
End if
End if



if then elseif

If (condition 1) then
statement 1

Elseif (condition 2) then
statement 2

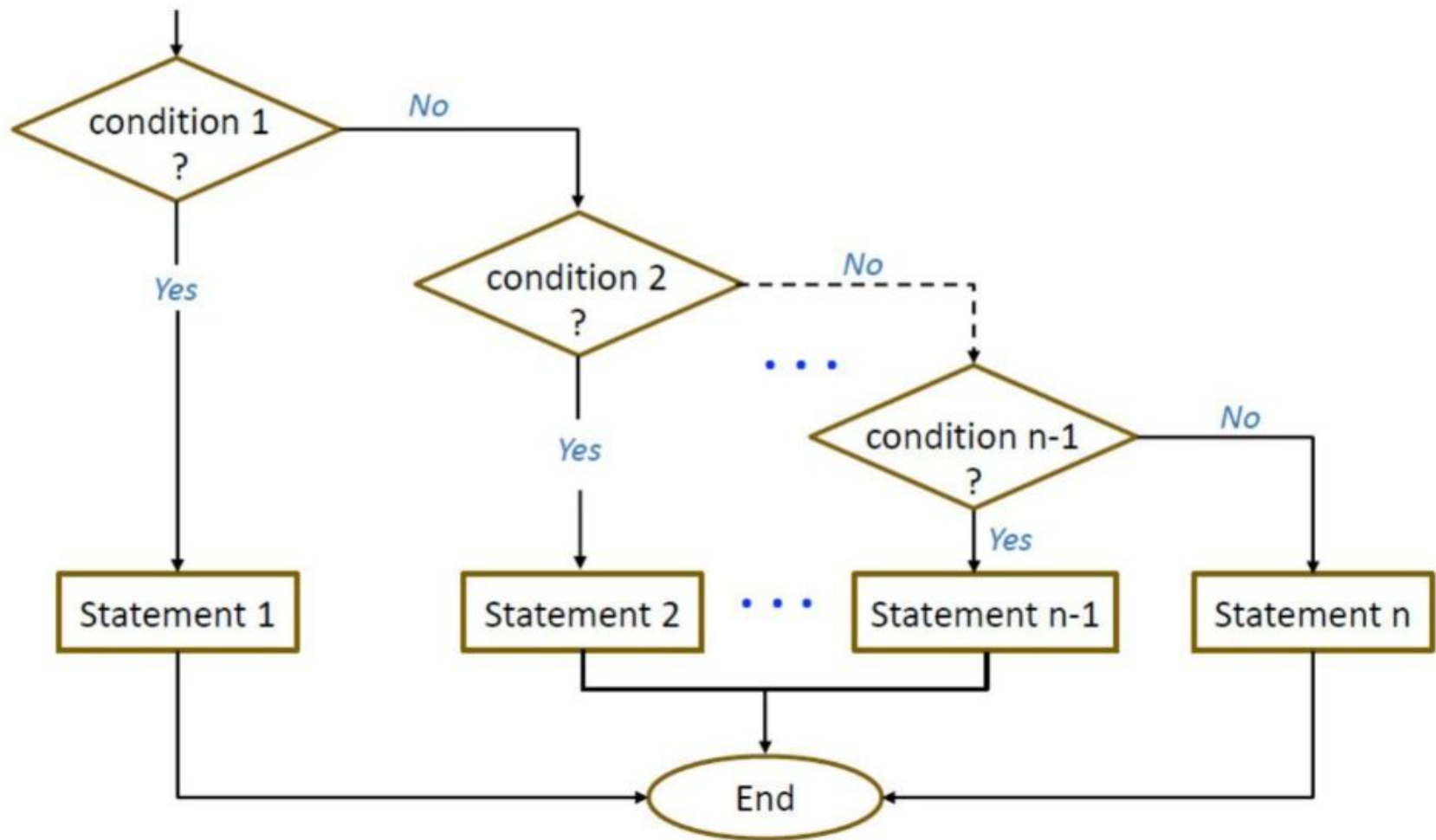
Elseif (condition 3) then
statement 3

Else
statement 4

End if

if ... then ... elseif – Flowchart

❖ **if then elseif** Flowchart



Condition – Example

- ✓ Write the pseudo-code to check if a student has passed the exam or failed. Then draw the equivalent flowchart?

Ans:

Step 1: Start

Step 2: Declare Result

Step 3: Read the student's grade

Step 4: *If* (Grade $\geq$ 50)

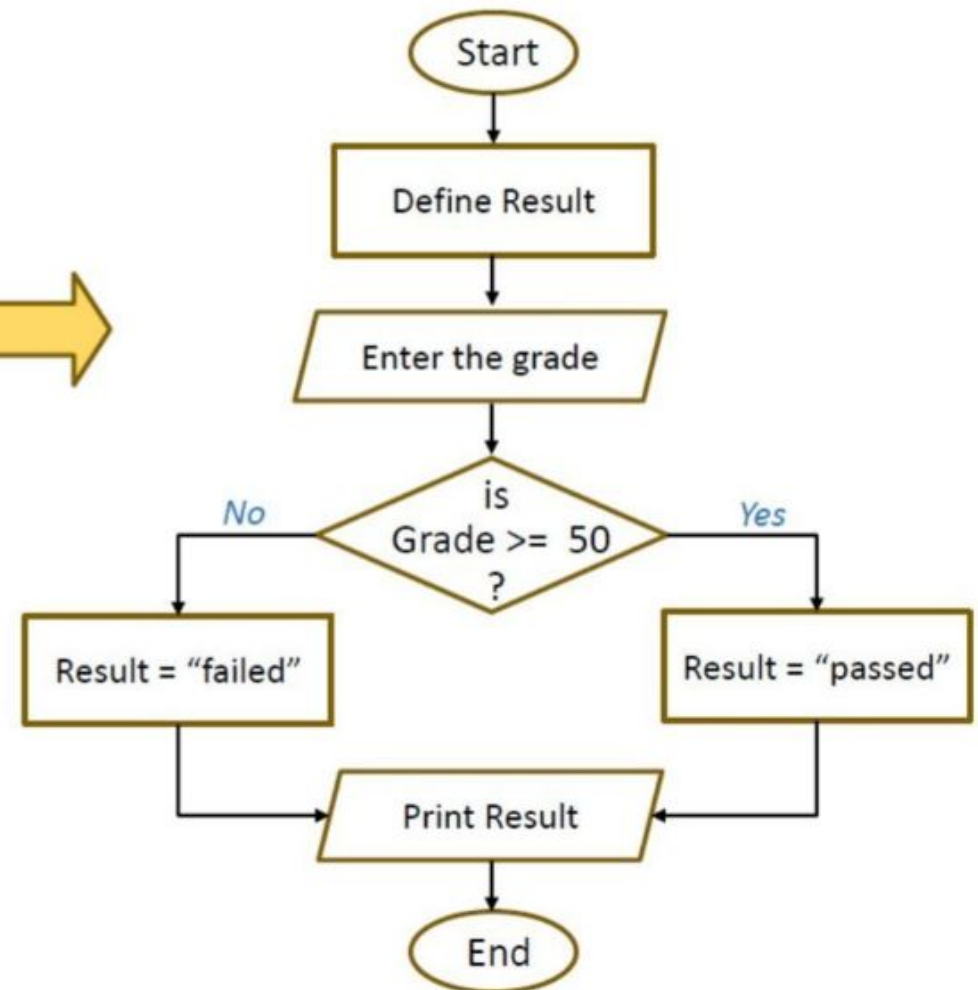
Result = "passed"

else

Result = "failed"

Step 5: Print result

Step 6: End



Classwork #1 – Student's Grade

✓ Send a message to a student depending on the obtained score?

Ans:

Step 1: Start

Step 2: Declare Result, message

Step 3: Read the student's grade

Step 4: *If* (Grade $\geq$ 50)

Result = "pass"

message = "Well Done!"

else

Result = "fail"

message = "See me!"

Step 5: Print result

Step 6: End

Draw the equivalent flowchart



Flowchart

Multiple Conditions – Example

- ✓ Is it possible to have multiple conditions in an algorithm?
- ❖ For example, print the result of a student according to the conditions listed in the table below?

Grade value	Printed result	التقدير
Grade < 50	Result = "Failed"	راسب
Grade < 60	Result = "Passed"	مقبول
Grade < 70	Result = "Medium"	متوسط
Grade < 80	Result = "Good"	جيد
Grade < 90	Result = "Very good"	جيد جداً
Grade > = 90	Result = "Excellent"	امتياز

Multiple Conditions – Example solution

Ans:

Step 1: Start

Step 2: Declare Grade, Result

Step 3: Read the student's grade

Step 4: *If* (Grade < 50)

Result = "failed"

else if (Grade < 60)

Result = "passed"

else if (Grade < 70)

Result = "medium"

else if (Grade < 80)

Result = "good"

تكملة الحل

else if (Grade < 90)

Result = "very good"

else

Result = "excellent"

Step 5: Print result

Step 6: End

Classwork #2 – Ticket price

- ✓ Write the pseudo-code algorithm to display the price of a cinema ticket as follow: (age < 12) then (price = 5K IQD), (age > 65) then (price = 7K IQD); otherwise (price = 10K IQD)?

Ans:

Step 1: Start

Step 2: Declare price

Step 3: Read the customer's Age

Step 4: *If* (Age < 12)

price = 5K

Else if (Age > 65)

price = 7K

else

price = 10K

Step 5: Display price

Step 6: End

Draw the equivalent flowchart



Flowchart

Iteration → Repetition

- ❖ Let's say we need to print all the numbers from 1 to 1000.
- ❖ Would we do this by sequencing?
- ❖ It is much better to do tasks like that using iteration

Repeat the operation or sequence of operations until the condition is met

Sequencing

Step 1: Start

Step 2: display 1

Step 3: display 2

Step 4: display 3

•
•
•

Step 1001: display 1000

Step 1002: END

Iteration

Step 1: Start

Step 2: Declare number = 1

Step 3: while the number $\leq$ 1000

{

display the number

number = number + 1 (increment the number)

}

Step 4: End

go to step 3
if
count $\leq$ 1000

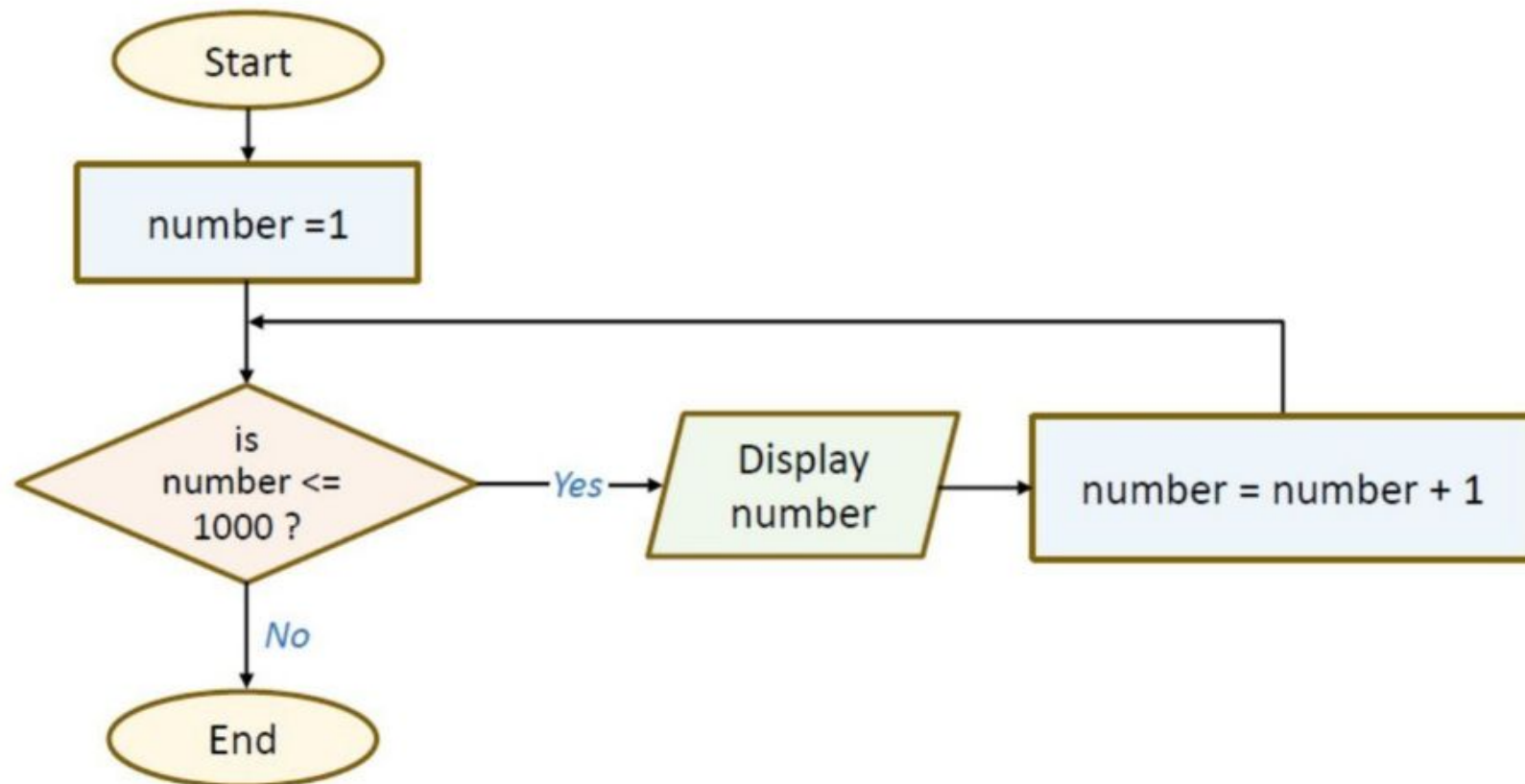
Loop



Iteration – printing the numbers from 1 to 1000

❖ Flowchart

➤ print the numbers from 1 to 1000



Iteration – Summing five numbers

✓ Write the pseudo-code for algorithm to sum five numbers. Then draw the equivalent flowchart?

Ans:

Step 1: Start

Step 2: Initialization

Initialize: $\text{sum} = 0$ and $\text{count} = 0$

Step 3: Enter n numbers

Step 4: Calculate sum and increment count

$\text{sum} = \text{sum} + n$

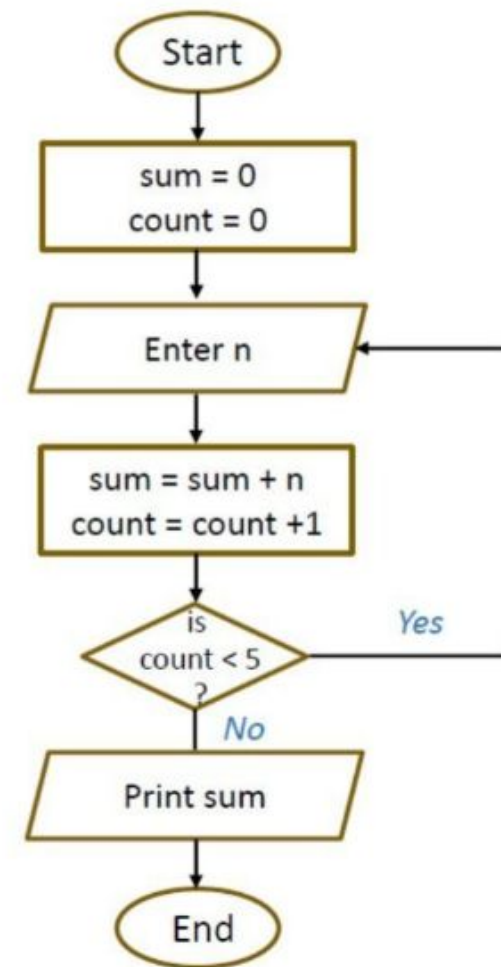
$\text{count} = \text{count} + 1$

Step 5: *while* ($\text{count} < 5$)

go to step 3

Step 6: Print sum

Step 7: End



Average of ten numbers – Example

 Write a pseudo-code algorithm to find the sum and average of ten numbers entered by the user. Then draw the equivalent flowchart.

Ans:

Algorithm

Step 1: Start

Step 2: Initialization

Initialize: sum = 0, Ave = 0, and count = 0

Step 3: while count < 10

{
 enter a **number**
 sum = sum + **number**
 count = count + 1
}

go to step 3 if count < 10

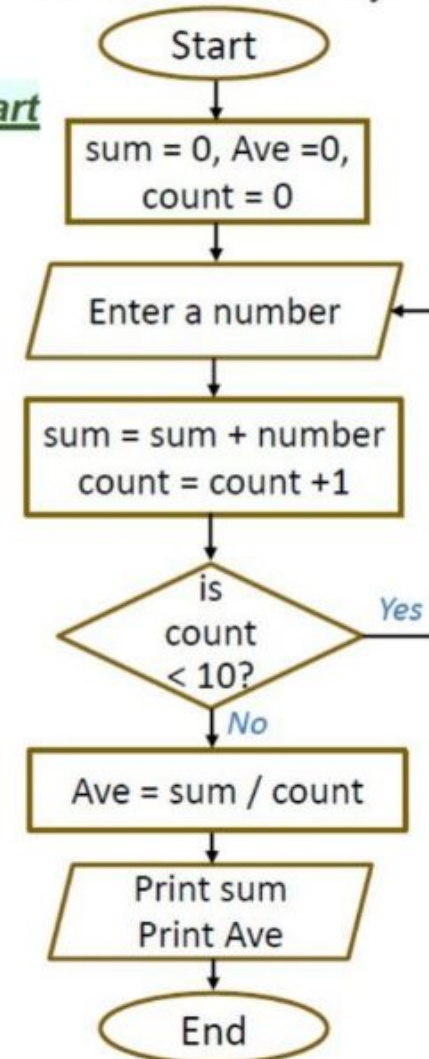
Step 4: Ave = sum / count

Step 6: Print sum

Print Ave

Step 7: End

Flowchart



Qualities of Good Algorithms

1. **Precision** – الدقة the steps of input and output should be precisely defined.
2. **Clarity** – الوضوح each step in the algorithm should be clear and unambiguous.
3. **Inputs/Outputs** – المدخلات والمخرجات has inputs that lead to the production of the output
4. **Effectiveness** – الفعالية Algorithms should solve problems in effective ways such that getting an
5. output from each step based on its inputs and outputs of the previous step
6. **Generality** – العمومية/الشمول the algorithm should be written in such a way that it can be used in
7. different programming languages.

THANK
YOU